Deconstructing Domain Shift in Diabetic Retinopathy Grading: Multi-Scale MIL with Consistent Ordinal Regression

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Abstract. Aggregate benchmark performance alone may not adequately characterize how automated five-class diabetic retinopathy (DR) grading behaves across heterogeneous clinical cohorts. We investigate cross-domain DR grading across six cohorts (APTOS 2019, DDR, DeepDRiD, IDRiD, Messidor-2, and EyePACS; N = 55,570 images from 35,474 patients) using patient-level six-fold Leave-One-Dataset-Out evaluation. We test three clinically motivated hypotheses linking native-resolution local Multiple Instance Learning (MIL) to fine-scale Grade 1 evidence, global relational modeling to spatially distributed Grade 3 evidence, and Cumulative Link Modeling (CLM) to the ordered ICDRS severity scale. Grade-specific sensitivity and referable-DR (RDR) AUROC and sensitivity at 95% specificity are used as clinically oriented cross-domain endpoints, with QWK and Within-1 accuracy as complementary aggregate and ordinal measures. Across held-out cohorts, QWK ranges from [X] to [Y], RDR AUROC from [X] to [Y], and RDR sensitivity at 95% specificity from [X]% to [Y]%. Grade-level and paired component analyses reveal target-dependent behavior at Grades 1 and 3 rather than uniform component benefits. In the controlled head comparison, CLM achieves [X]% Within-1 accuracy, whereas [HEAD] achieves the highest QWK and [HEAD] the highest RDR AUROC, indicating metric-specific trade-offs among severity proximity, aggregate agreement, and screening discrimination. These results show that comparable aggregate performance can coexist with substantially different clinically relevant behavior, supporting grade-aware evaluation of cross-domain DR grading.

Keywords: Diabetic Retinopathy · Multi-Scale Deep Learning · Multiple Instance Learning (MIL) · Cumulative Link Model (CLM) · Domain Generalization · Ordinal Regression

1 Introduction

Diabetic retinopathy (DR) is a progressive microvascular complication of diabetes mellitus and a leading cause of preventable blindness worldwide [28].

Automated grading from color fundus photographs enables scalable screening, where clinical assessment follows the International Clinical Diabetic Retinopathy Disease Severity Scale (ICDRS), an ordered five-grade severity scale [25]. In screening workflows, errors between distant severity grades can carry greater triage implications than adjacent-grade errors: confusing Grade 3 with Grade 0 can under-triage severe disease, whereas confusing Grade 1 with Grade 0 represents a smaller severity discrepancy. At the same time, aggregate metrics such as Quadratic Weighted Kappa (QWK) and overall accuracy summarize performance across disease stages and may not adequately characterize how performance varies across specific clinical grades under distribution shift.

Deep learning models can achieve strong in-domain performance on closed benchmarks, with reported QWK values exceeding 0.90 [7]. However, external validation reveals substantial sensitivity to differences in imaging devices, acquisition conditions, and patient populations [18,27]; referable-DR sensitivity has also been reported to decline on unseen cohorts [13]. While domain generalization (DG) benchmarks have emerged to assess zero-shot transfer on unseen retinal datasets [9], evaluation remains largely centered on aggregate cohort-level metrics. Such comparisons can be difficult to interpret in isolation because target cohorts may also differ in grade prevalence, annotation practices, and intrinsic difficulty. Consequently, an important question remains: *To what extent do aggregate cross-cohort metrics adequately characterize generalization across individual clinical grades, and which modeling mechanisms are associated with the observed grade-specific performance patterns?* We therefore characterize cross-domain behavior at both the clinical-grade and architectural-component levels rather than reducing generalization to a single aggregate outcome.

The ICDRS criteria provide a clinical basis for three pre-specified modeling hypotheses. Grade 1 is defined by microaneurysms only, making fine-scale lesion evidence particularly relevant [25]. This motivates RQ1: *How does Grade 1 behave under held-out-domain evaluation, and is native-resolution local Multiple Instance Learning (MIL) associated with a consistent Grade 1-specific change across unseen cohorts?* Grade 3 is defined by the multi-quadrant "4-2-1 rule," requiring evidence distributed across spatially separated retinal regions [25]. This motivates RQ2: *How does Grade 3 behave across unseen cohorts, and is global relational and quadrant-level context associated with a consistent Grade 3-specific change?* Finally, because the ICDRS grades form an ordered severity scale ($0 < 1 < 2 < 3 < 4$), RQ3 asks: *Does explicit ordinal modeling alter severity-consistent prediction under held-out-domain evaluation relative to nominal and alternative ordinal formulations?*

We evaluate these questions using patient-level six-fold Leave-One-Dataset-Out (LODO) evaluation across six heterogeneous retinal cohorts. Rather than assuming a uniform cross-domain performance decrease or interpreting raw source–target QWK differences as direct domain-shift penalties, the study decomposes generalization by clinical grade and matched component comparison, with ordinal consistency and referable-DR screening providing complementary clinical views.

2 Related Work

2.1 Deep Learning for DR Grading and Domain Generalization

Automated grading of diabetic retinopathy (DR) has advanced from binary referable-disease screening [11,22] to five-class staging under the International Clinical Diabetic Retinopathy Disease Severity Scale (ICDRS) [20,7]. Although contemporary deep networks achieve strong performance on established benchmarks, external validation has revealed substantial cross-cohort performance variability [3,18]. Such variability has been associated with differences in imaging devices, illumination, field of view, and patient populations [27], and clinically relevant screening sensitivity has also been reported to change substantially on unseen cohorts [13].

Domain generalization (DG) seeks representations that transfer zero-shot to unseen targets without requiring target-domain data during training [10,9]. Galappaththige et al. [9] established a systematic fundus DG benchmark across single- and multi-source settings. However, cross-domain evaluation remains primarily centered on cohort-level aggregate metrics such as QWK, overall accuracy, or AUC, leaving grade-specific behavior less systematically characterized.

2.2 Multi-Scale and Multi-Region Architectures for Retinal Pathology

Retinal pathology varies substantially in spatial scale and distribution, motivating architectures that combine fine-grained visual information with broader retinal context [2,16]. Conventional whole-image pipelines operate on resized fundus photographs, whereas multi-scale and patch-based approaches preserve higher-resolution local information. Multiple Instance Learning (MIL) provides a general framework for aggregating instance-level representations into image-level predictions [15].

Complementary approaches model information across spatially separated retinal regions. Quadrant-based methods explicitly preserve regional organization [5], while global-local architectures combine localized representations with broader contextual information through attention or relational modeling [16]. These approaches provide complementary strategies for representing localized pathology and distributed retinal context. However, whether local and global modeling strategies produce consistent, grade-specific effects across unseen retinal cohorts remains less systematically characterized.

2.3 Ordinal Classification and Cumulative Link Modeling in Medical Imaging

Clinical DR staging follows an ordered severity scale, $\mathcal{Y} = \{0 < 1 < 2 < 3 < 4\}$ [25]. Standard nominal Softmax classification treats grades as unordered categories and does not explicitly encode the severity distance between adjacent and distant classification errors [17,4]. Continuous regression introduces a notion of

distance but imposes metric spacing among grade labels that is not explicitly defined by the ICDRS scale [19].

Ordinal formulations instead model class ordering directly. CORAL decomposes multi-class prediction into $K-1$ rank-consistent binary tasks with shared parameters [6], while CORN models conditional ordinal probabilities through sequential binary classifiers [21]. Cumulative Link Models (CLMs) map representations to a latent severity axis partitioned by ordered cutpoints and can be combined with weighted-kappa objectives [1,23]. Although ordinal prediction has been studied extensively for medical image grading, its behavior under systematic multi-cohort held-out-domain evaluation remains less established.

Prior work establishes the relevance of cross-domain evaluation, local/global retinal modeling, and ordinal prediction, yet their integration remains limited. We therefore use a common multi-cohort LODO protocol to examine grade-specific behavior and the consistency of clinically motivated mechanisms beyond aggregate performance.

3 Methodology

3.1 Framework Overview

We formulate diabetic retinopathy (DR) grading as a five-class ordinal prediction problem over color fundus photographs, with labels $y \in \{0, 1, 2, 3, 4\}$ following the ICDRS severity scale. As illustrated in Fig. 1, the framework combines a fixed-resolution global context stream with a native-resolution local MIL stream, whose representations are fused and mapped to ordinal severity using a Cumulative Link Model (CLM).

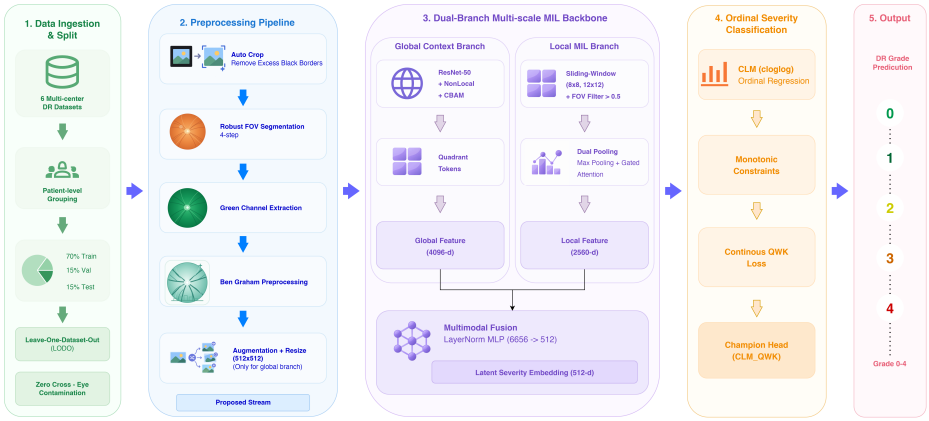


Fig. 1. Overview of the multi-scale ordinal MIL framework for grade-aware cross-domain DR analysis.

3.2 Cohort Harmonization & Patient-Level Partitioning

We construct a harmonized five-class DR cohort from six heterogeneous fundus datasets: APTOS 2019, DDR, DeepDRiD, IDRiD, Messidor-2, and EyePACS. The cohort composition, imaging characteristics, patient counts, and grading protocols are summarized in Table 1

Table 1. Composition, imaging characteristics, and clinical severity distributions across the six harmonized fundus cohorts ($N = 55{,}570$ images, 35,474 patients).

Cohort	Country	Device / FOV	Images (Patients)	Grade Distribution (%) [G_0 / G_1 / G_2 / G_3 / G_4]	Grading Protocol
APTOS 2019	India	Zeiss, Topcon (45°)	3,662 (3,662)	49.3 / 10.1 / 27.3 / 5.3 / 8.1	Single-grader ICDRS
DDR	China	Canon CR-2, Topcon (45°)	12,522 (12,522)	50.0 / 5.0 / 35.8 / 1.9 / 7.3	Multi-expert consensus
DeepDRiD	China	Topcon TRC-NW400 (45°)	2,000 (500)	34.0 / 17.8 / 19.6 / 19.6 / 9.0	2-field paired stereoscopic
IDRiD	India	Kowa VX-10α (50°)	516 (516)	32.6 / 4.8 / 32.6 / 18.0 / 12.0	Multi-expert + lesion masks
Messidor-2	France	Topcon TRC NW6 (45°)	1,744 (711)	58.3 / 15.5 / 19.9 / 4.3 / 2.0	Multi-grader consensus
EyePACS	USA	Centervue, Canon (45°)	35,126 (17,563)	73.5 / 7.0 / 15.1 / 2.5 / 2.0	Real-world tele-screening
Total / Harmonized	Multi	Diverse Multi-Vendor (45°–50°)	**55,570 (35,474)**	**64.3 / 7.4 / 21.0 / 3.4 / 3.9**	**ICDRS 5-Grade Staging**

To prevent patient-level leakage, each patient is identified using the compound key `dataset_key::patient_id`. All partitions are generated at the patient level, preventing paired-eye or repeated-image records from being distributed across training and evaluation partitions.

3.3 Preprocessing and Optical FOV Segmentation

To harmonize heterogeneous fundus images while preserving native spatial resolution, each image is tightly cropped to its retinal region using a grayscale-based non-black-region mask. A binary optical field-of-view (FOV) mask is then derived from the green channel using Otsu thresholding, morphological operations, largest-connected-component selection, and inward erosion. The mask is used to exclude patches dominated by non-retinal regions during local instance construction.

For model input, the RGB image is processed by bilateral filtering followed by Ben Graham-style [12]:

$$I_{\text{enh}} = 4I_G - 4G_{\sigma=30}(I_G) + 128, \tag{1}$$

where I_G denotes the filtered green channel and $G_{\sigma=30}(\cdot)$ denotes Gaussian background estimation. The enhanced channel is replicated across three channels and used as the common representation for both streams.

3.4 Global Context Stream

The global context stream extracts a fixed-resolution representation of the entire fundus image while explicitly modeling relationships among spatially separated retinal regions. The preprocessed image is resized to $512 \times 512 \times 3$ and encoded by a ResNet-50 backbone [14], producing a Layer-4 feature map $\mathbf{X}_4 \in \mathbb{R}^{2048 \times 16 \times 16}$.

A Non-Local Relational Block [24] is applied to model dependencies between spatially separated feature locations. The resulting feature map is subsequently refined using channel-wise and spatial-wise CBAM attention [26].

To retain coarse spatial organization relevant to multi-quadrant assessment, the attended feature map is partitioned into four non-overlapping spatial quadrants. Each 8×8 quadrant is pooled into a token, augmented with a learned positional embedding, and processed jointly by a one-layer Transformer encoder:

$$\mathbf{F}_{\text{Quadrant}} = \text{TransformerEncoder}\left([\mathbf{t}_m + \mathbf{p}_m]_{m=1}^{4}\right) \in \mathbb{R}^{2048}. \tag{2}$$

In parallel, global average pooling (GAP) is applied to produce a whole-image summary of the refined feature map. The final global representation is $\mathbf{F}_{\text{Global}} = [\mathbf{F}_{\text{GAP}} \| \mathbf{F}_{\text{Quadrant}}] \in \mathbb{R}^{4096}$.

3.5 Native-Resolution Local Multiple Instance Learning (MIL) Stream

The local stream operates at native spatial resolution to retain fine-scale image information. From the native-resolution preprocessed image, the stream extracts overlapping $128 \times 128 \times 3$ patches using a sliding window with stride $s = 96$ pixels, corresponding to 25% spatial overlap. Patches with less than 50% FOV coverage are discarded to exclude patches dominated by non-retinal regions.

Each valid patch is encoded by an EfficientNet-B0 backbone, producing an instance embedding $\mathbf{h}_k \in \mathbb{R}^{1280}$. The resulting variable sized bag $\{\mathbf{h}_1, \ldots, \mathbf{h}_M\}$ is aggregated using complementary max pooling and gated attention [15]. Max pooling preserves the strongest instance-level responses, $\mathbf{F}_{\text{Max}} = \max_{k \in \{1,\ldots,M\}} \mathbf{h}_k \in \mathbb{R}^{1280}$.

For content-adaptive aggregation, gated attention assigns an instance-level score $\mathbf{w}_a^\top (\tanh(\mathbf{V}\mathbf{h}_k) \odot \sigma(\mathbf{U}\mathbf{h}_k))$ following [15].

The two complementary summaries are concatenated to obtain $\mathbf{F}_{\text{Local}} = [\mathbf{F}_{\text{Max}} \| \mathbf{F}_{\text{Att}}] \in \mathbb{R}^{2560}$.

During training, $M \in [32, 64]$ valid patches are stochastically sampled per image to control computational cost and instance diversity. During evaluation, a dense sliding-window sweep is used, with up to 256 valid patches per image.

3.6 Multi-Scale Feature Fusion and Ordinal Prediction

The global and local representations are concatenated to form the joint multi-scale descriptor $\mathbf{F}_{\text{Fused}} = [\mathbf{F}_{\text{Global}} \| \mathbf{F}_{\text{Local}}] \in \mathbb{R}^{6656}$. A two-layer MLP with LayerNorm, Mish activation and dropout maps $\mathbf{F}_{\text{Fused}}$ to $\mathbf{z} \in \mathbb{R}^{512}$.

The latent representation is projected onto a continuous one-dimensional severity axis, $s(\mathbf{x}) = \mathbf{w}^\top \mathbf{z}$. The latent severity axis is partitioned into five ordered intervals using four strictly increasing cutpoints. Their increments are parameterized through a positive softplus transformation:

$$\theta_0 \in \mathbb{R}, \qquad \theta_{k+1} = \theta_k + \text{softplus}(\alpha_k) + 10^{-4}, \qquad k \in \{0, 1, 2\}, \tag{3}$$

where α_k are unconstrained learnable parameters.

The cumulative probabilities are parameterized using the complementary log-log link:

$$P(Y \leq k \mid \mathbf{x}) = 1 - \exp\left[-\exp\left(\theta_k - s(\mathbf{x})\right)\right], \qquad k \in \{0, 1, 2, 3\}. \tag{4}$$

The five class probabilities are obtained from adjacent cumulative probabilities:

$$\begin{aligned} &p_0 = P(Y \leq 0), \qquad p_k = P(Y \leq k) - P(Y \leq k-1), \quad k = 1, \ldots, 3, \\ &p_4 = 1 - P(Y \leq 3). \end{aligned} \tag{5}$$

3.7 Training Objective

The model is optimized using a composite objective that combines the interval negative log-likelihood (NLL) of the ordinal probabilities with a differentiable QWK-based disagreement loss, $\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{NLL}} + \lambda(t)\mathcal{L}_{\text{QWK}}$.

The QWK-based term λ warms linearly from 0 to $\lambda_{\text{target}} = 0.4$ over the first 15 epochs.

4 Experimental Protocol

4.1 Leave-One-Dataset-Out (LODO) Evaluation

We perform six-fold Leave-One-Dataset-Out (LODO) evaluation across the six harmonized cohorts introduced in Section 3.2. In fold k, five cohorts form the development pool, while the sixth cohort is held out entirely for zero-shot evaluation:

$$\mathcal{D}^{(k)}_{\text{Train}} = \bigcup_{j \neq k} \mathcal{D}_{j,\text{train}}, \qquad \mathcal{D}^{(k)}_{\text{Val}} = \bigcup_{j \neq k} \mathcal{D}_{j,\text{val}}, \qquad \mathcal{D}^{(k)}_{\text{OOD}} = \mathcal{D}_{k,\text{test}}. \tag{6}$$

The training partitions of the five source cohorts are used for model optimization, while their validation partitions are used for checkpoint selection and hyperparameter selection. The corresponding source test partitions remain unseen during model selection and are used only for in-domain evaluation. The sixth cohort is used exclusively as the zero-shot out-of-domain (OOD) test set.

For each fold, aggregate QWK is reported separately for the pooled source-domain test partitions and the held-out target cohort. Because these evaluations may differ in grade distribution and cohort-specific difficulty, their raw difference is treated only as a descriptive cross-cohort comparison rather than as a direct estimate of domain-shift severity. Cross-domain behavior is instead characterized primarily through grade-specific sensitivity and referable-DR screening performance on the held-out cohort.

4.2 Controlled Component Variants

We evaluate a stepwise architecture sequence under the same six-fold LODO protocol. The variants are defined as follows: V_0 uses a global CNN with Softmax; V_1 adds native-resolution local MIL; V_2 further adds global relational context; and V_3 replaces the Softmax head with the CLM-based ordinal head.

The successive comparisons $V_1 - V_0$, $V_2 - V_1$, and $V_3 - V_2$ are interpreted as incremental component comparisons under a fixed experimental protocol. In particular, they assess the changes associated with adding local-resolution modeling, global relational context, and ordinal prediction, respectively, rather than claiming causal isolation of individual operations within each component.

4.3 Ordinal Head Benchmark

To compare alternative output formulations under a fixed representation pipeline, we evaluate five prediction heads using the same single-branch ResNet-50 backbone at 512×512 resolution and the same data partitions and optimization protocol: Softmax-CE, Softmax-QWK, CORAL [6], CORN [21], and CLM-QWK with a complementary log-log link and monotonic cutpoints [23].

4.4 Training and Model Selection Protocol

All component variants and output-head configurations are trained under a common optimization protocol to ensure comparability. Models are optimized using AdamW with weight decay 10^{-4}. The initial learning rates are 10^{-4} for newly initialized parameters and 10^{-5} for pretrained backbone parameters. A cosine-annealing schedule is applied over 100 training epochs with a minimum learning rate of 10^{-6}. The batch size is 16.

Checkpoint selection is performed exclusively on the pooled validation partitions of the five source cohorts. Early stopping with a patience of 15 epochs is based on validation QWK, and the best checkpoint is restored for subsequent evaluation. Each configuration is trained once per LODO fold. For the pre-specified component comparisons, matched fold-level outcomes are analyzed using two-sided exact Wilcoxon signed-rank tests ($n = 6$). Because the small number of folds yields coarse exact p-values, statistical results are interpreted together with the median paired difference, interquartile range, and direction of change across cohorts rather than as stand-alone evidence.

For configurations incorporating the differentiable QWK-based disagreement loss, the loss weight is linearly warmed from zero to $\lambda_{\text{target}} = 0.4$ over the first 15 epochs. Class weighting is applied during optimization to mitigate training-set imbalance, while validation and test partitions retain their natural class distributions.

4.5 Evaluation Metrics

Cross-domain behavior is evaluated using complementary clinical-grade, screening, and aggregate measures. Grade-specific sensitivity provides the primary

evidence for the Grade 1 and Grade 3 analyses, while referable-DR (RDR; Grade ≥ 2) performance is assessed using AUROC and sensitivity at 95% specificity. For each LODO fold, the operating threshold for 95% specificity is selected exclusively on the pooled source validation partitions and fixed before evaluation on the held-out cohort. QWK is reported as a complementary measure of overall five-class ordinal agreement [8], while Within-1 accuracy characterizes the severity proximity of prediction errors. Together, these metrics are used to distinguish aggregate performance from grade-specific and clinically oriented cross-domain behavior.

4.6 Quantitative Explainable AI Evaluation

To assess spatial alignment between model saliency and expert-annotated retinal lesions, we evaluate saliency localization on the available IDRiD subset ($N = 81$ images), covering microaneurysms, hemorrhages, hard exudates, and soft exudates. Saliency maps from the global and local streams are mapped to the original image coordinates using the recorded preprocessing transformations. We report **Pointing Game accuracy**, **Area-Matched Saliency Recall**, and **pixel-level AUROC**, measuring respectively whether the maximum-saliency location falls within a lesion region, the fraction of annotated lesion pixels covered by an area-matched salient region, and pixel-wise discrimination between lesion and non-lesion retinal regions within the FOV. This analysis is treated as supporting anatomical localization evidence rather than a direct measure of predictive performance or causal correspondence.

5 Results and Discussion

5.1 Cross-Domain Performance Beyond Aggregate QWK

We first ask whether aggregate performance alone provides an adequate account of model behavior across unseen cohorts. Under the baseline V_0, held-out QWK ranges from 0.533 on EyePACS to 0.818 on APTOS 2019. In contrast, Grade 1 sensitivity ranges from 7.0% to 88.0%, Grade 3 sensitivity from 29.0% to 65.8%, and RDR sensitivity at 95% specificity from 51.4% to 82.4%.

The cohort-level results show that similar or high aggregate agreement does not uniquely determine performance at a particular clinical grade. For example, APTOS reaches the highest baseline QWK (0.818) while Grade 1 sensitivity is only 7.0%, whereas IDRiD obtains a lower QWK (0.714) but substantially higher Grade 1 sensitivity (88.0%). The same distinction is apparent across the component variants: mean OOD QWK remains nearly unchanged for V_0, V_1, and V_2 (0.700, 0.700, and 0.698), although their grade-level sensitivity profiles differ substantially.

These observations establish the motivation for the subsequent hypothesis tests: component effects are evaluated at the clinical grades that motivated their design rather than inferred from aggregate QWK alone.

Table 2. Six-fold Leave-One-Dataset-Out (LODO) generalization benchmark. Each cell reports ID QWK / OOD QWK, followed by the corresponding generalization gap $\Delta_{\mathrm{QWK}} = \mathrm{QWK}_{\mathrm{ID}} - \mathrm{QWK}_{\mathrm{OOD}}$.

Held-out Cohort	V_0	V_1	V_2	V_3
APTOS 2019	0.70 / 0.82 (-0.12)	0.69 / 0.80 (-0.11)	0.70 / 0.80 (-0.10)	[X] / [X] ([X])
IDRiD	0.75 / 0.71 (0.04)	0.74 / 0.82 (-0.08)	0.75 / 0.82 (-0.07)	[X] / [X] ([X])
Messidor-2	0.75 / 0.65 (0.10)	0.77 / 0.70 (0.07)	0.71 / 0.69 (0.02)	[X] / [X] ([X])
DDR	0.76 / 0.72 (0.04)	0.76 / 0.60 (0.16)	0.75 / 0.59 (0.16)	[X] / [X] ([X])
EyePACs	0.83 / 0.53 (0.30)	0.82 / 0.55 (0.27)	0.81 / 0.51 (0.30)	[X] / [X] ([X])
DeepDRiD	0.72 / 0.76 (-0.04)	0.72 / 0.74 (-0.02)	0.72 / 0.78 (-0.06)	[X] / [X] ([X])
Mean	**[X]**	**[X]**	**[X]**	**[X]**

5.2 Native-Resolution Local MIL and Grade 1 Sensitivity

RQ1 tests whether adding native-resolution local MIL produces a consistent Grade 1-specific effect across unseen cohorts. Averaged across the six targets, Grade 1 sensitivity changes from 39.5% under V_0 to 31.6% under V_1. This mean, however, masks pronounced cross-cohort heterogeneity: Grade 1 sensitivity is higher under V_1 in four of six targets, but large negative changes on IDRiD and Messidor-2 dominate the fold average.

The median paired change is +7.8 percentage points (IQR: −28.5 to +12.6 pp), whereas the mean change is −7.9 pp. The two-sided exact Wilcoxon signed-rank test does not indicate a consistent fold-level effect ($W = 10$, $p = 1.000$). Moreover, the transition is not selectively favorable to Grade 1: Grade 2 sensitivity increases by 10.7 pp on average, whereas Grade 1 decreases by 7.9 pp.

Thus, the current six-cohort evidence does not support a consistent Grade 1-specific benefit from native-resolution local MIL. Instead, the local component exhibits target-dependent effects, indicating that preserving native-resolution evidence alone is insufficient to guarantee uniform mild-DR generalization across heterogeneous unseen cohorts.

Table 3. Mean held-out performance across six LODO targets for the component variants.

Metric	V_0	V_1	V_2	V_3
Grade 0 Sens.	69.4%	71.0%	62.0%	[X]%
Grade 1 Sens.	39.5%	31.6%	45.5%	[X]%
Grade 2 Sens.	40.3%	51.0%	44.6%	[X]%
Grade 3 Sens.	45.6%	43.9%	43.8%	[X]%
Grade 4 Sens.	59.1%	56.9%	57.3%	[X]%
Within-1 Acc.	90.1%	89.3%	90.7%	[X]%
RDR Sens.@95%	72.0%	70.3%	73.0%	[X]%
Mean QWK	0.700	0.700	0.698	[X]

5.3 Global Relational Context and Grade 3 Sensitivity

RQ2 examines whether the addition of global relational and quadrant-level context is associated with a consistent Grade 3-specific change. Mean Grade 3 sensitivity is essentially unchanged between V_1 and V_2 (43.85% versus 43.84%). The paired effect is positive on two held-out cohorts and negative on four, yielding a median change of -3.7 percentage points (IQR: -6.4 to $+5.6$ pp). The exact Wilcoxon test likewise provides no evidence of a consistent fold-level effect ($W = 10$, $p = 1.000$).

The broader grade profile also argues against a selectively Grade 3-specific response. Over the same transition, Grade 1 sensitivity increases by 13.9 pp on average, whereas Grade 0 decreases by 9.0 pp and Grade 3 remains nearly unchanged. The added global context redistributes grade-level behavior without producing the preferential Grade 3 improvement hypothesized in RQ2.

These results do not provide clear support for H2 under the present LODO setting. Because $V_1 \rightarrow V_2$ jointly introduces several global-context operations, the finding is interpreted at the component level and does not isolate the contribution of any individual Non-Local, CBAM, or quadrant-token operation.

5.4 Ordinal Modeling and Severity Consistency

RQ3 asks whether explicitly modeling the ordered ICDRS scale changes the severity structure of prediction errors. The controlled output-head benchmark reveals a metric-dependent trade-off rather than uniform superiority of a single formulation. CORAL achieves the highest QWK (0.765), whereas CLM obtains the highest Within-1 accuracy (95.01%). CORN yields the highest RDR AUROC (0.933), while Softmax-QWK provides the highest sensitivity at 95% specificity (72.39%).

Relative to nominal Softmax-CE, CLM increases Within-1 accuracy from 93.20% to 95.01% and QWK from 0.696 to 0.715, but its RDR AUROC decreases from 0.925 to 0.916 and sensitivity at 95% specificity from 71.04% to 66.16%. The ordinal formulation appears to favor severity proximity without yielding uniformly stronger aggregate agreement or screening discrimination.

Table 4. Controlled comparison of output formulations under the standardized ResNet-50 benchmark. Best values are highlighted per metric.

Head	QWK	Within-1	RDR Sens.@95%	RDR AUROC
$\text{Softmax}_{\text{CE}}$	0.696	93.20%	71.04%	0.925
$\text{Softmax}_{\text{QWK}}$	0.724	93.92%	**72.39%**	0.929
CORAL	**0.765**	93.61%	70.38%	0.927
CORN	0.678	94.34%	71.56%	**0.933**
CLM_{QWK}	0.715	**95.01%**	66.16%	0.916

5.5 Clinical Implications for Referable-DR Screening

We next examine whether the grade-level component effects translate to the referable-DR decision boundary. Mean held-out sensitivity at the validation-selected 95%-specificity operating point is 72.0% for V_0, 70.3% for V_1, and 73.0% for V_2. The transition from V_0 to V_1 lowers sensitivity on five of six targets, whereas $V_1 \rightarrow V_2$ produces mixed directions across the six cohorts.

Thus, the component sequence does not yield a monotonic improvement in screening performance despite substantial changes in the underlying grade-specific sensitivity profile. This divergence further supports treating fine-grained grading behavior and binary RDR discrimination as complementary views of cross-domain performance rather than assuming that improvement in one necessarily implies improvement in the other.

5.6 Lesion-Level Localization Evidence

We further examine whether the spatial evidence used by the global and local branches overlaps with expert-annotated retinal lesions. On the IDRiD subset with expert lesion masks ($N = 81$), the Local Attention-MIL stream achieves a Pointing Game accuracy of $[X]$%, compared with $[X]$% for global Grad-CAM. The corresponding Area-Matched Recall and pixel-level AUROC are $[X]$ and $[X]$ for the local stream, compared with $[X]$ and $[X]$ for the global stream.

Figure **??** provides qualitative examples of the spatial correspondence between the fundus image, global Grad-CAM, local Attention-MIL saliency, and expert lesion masks. The local saliency maps exhibit more spatially concentrated activation around annotated lesions in representative cases, whereas global Grad-CAM produces broader activation regions.

These results provide supporting evidence that the native-resolution local stream can preserve spatially localized evidence relevant to retinal lesions. They are interpreted as localization evidence rather than as a direct causal explanation of the model's predictions.

5.7 Limitations

This study examines whether aggregate performance alone adequately characterizes cross-domain five-class diabetic retinopathy grading. Across six patient-level LODO cohorts, similar aggregate agreement can coexist with substantially different grade-specific and clinically oriented behavior. Paired component analyses further indicate that clinically motivated local and global mechanisms can have target-dependent rather than uniformly grade-specific effects, while alternative output formulations exhibit different trade-offs between aggregate agreement, severity proximity, and referable-DR discrimination.

These findings support evaluating cross-domain DR grading through complementary aggregate, grade-specific, ordinal, and screening endpoints rather than reducing generalization to a single benchmark score or raw source–target difference. Such decomposition provides a more clinically informative view of how modeling choices behave across heterogeneous unseen cohorts.

Code and Data Availability

We use six publicly available diabetic retinopathy cohorts, including APTOS 2019, DDR, DeepDRiD, IDRiD, Messidor-2, and EyePACS, accessed through their respective official or authorized distribution channels. The datasets are used for five-class diabetic retinopathy grading under patient-level partitioning and six-fold Leave-One-Dataset-Out (LODO) evaluation. Dataset access is subject to the terms and conditions of their respective providers.

Code, model configurations, preprocessing and audit scripts, patient-level split manifests, LODO split configurations, and evaluation scripts are archived at 10.5281/zenodo.22834483. The original fundus images are not redistributed.

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